KAIZEN MATHS

Lesson starter

Limits and First Principles

Parameters and Tangents

Name _____

S1

Find the parameter value that makes each function continuous.

Find k so that $f(x) = \{3x - 1, \quad x2$ is continuous at $x = 2$.

S2

Find the parameter so the limit exists, then evaluate the limit.

Find k so that $\lim_{x \to -2} \dfrac{x^2 + 6x + k}{x + 2}$ exists, then evaluate the limit.

S3

Find each tangent equation using a gradient from first principles.

Use first principles to find the equation of the tangent to $y = 2x^2 + 2$ at $x = -2$.

S4

Find the parameter value that makes each function continuous.

Find k so that $f(x) = \{2x - 3, \quad x3$ is continuous at $x = 3$.

Show your working

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KAIZEN MATHS

Exit ticket

Limits and First Principles

Parameters and Tangents

Name _____

EA1

Find the parameter so the limit exists, then evaluate the limit.

Find k so that $\lim_{x \to 2} \dfrac{x^2 + 2x + k}{x - 2}$ exists, then evaluate the limit.

EA2

Find each tangent equation using a gradient from first principles.

Use first principles to find the equation of the tangent to $y = x^2 + 4$ at $x = 4$.

Show your working

KAIZEN MATHS

Exit ticket

Limits and First Principles

Parameters and Tangents

Name _____

EB1

Find the parameter value that makes each function continuous.

Find k so that $f(x) = \{2x + 1, \quad x0$ is continuous at $x = 0$.

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KAIZEN MATHS

Exit ticket

Limits and First Principles

Parameters and Tangents

Name _____

EA1

Find the parameter so the limit exists, then evaluate the limit.

Find k so that $\lim_{x \to 2} \dfrac{x^2 + 2x + k}{x - 2}$ exists, then evaluate the limit.

EA2

Find each tangent equation using a gradient from first principles.

Use first principles to find the equation of the tangent to $y = x^2 + 4$ at $x = 4$.

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KAIZEN MATHS

Exit ticket

Limits and First Principles

Parameters and Tangents

Name _____

EB1

Find the parameter value that makes each function continuous.

Find k so that $f(x) = \{2x + 1, \quad x0$ is continuous at $x = 0$.

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Teacher answers & working

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S1

Find the parameter value that makes each function continuous.

Find k so that $f(x) = \{3x - 1, \quad x2$ is continuous at $x = 2$.

Answer:

$k = 5$

Working:

For continuity, the left-hand limit, right-hand limit, and function value must all be equal.

$$\lim_{x \to 2^-} f(x) = 3(2) - 1$$
$$= 5$$
$$\lim_{x \to 2^+} f(x) = -(2) + 7$$
$$= 5$$

The two one-sided limits are equal, so $\lim_{x \to 2} f(x) = 5$.

Set the function value equal to the limit: $k = 5$.

S2

Find the parameter so the limit exists, then evaluate the limit.

Find k so that $\lim_{x \to -2} \dfrac{x^2 + 6x + k}{x + 2}$ exists, then evaluate the limit.

Answer:

$k = 8, \ \lim = 2$

Working:

For a finite limit, the numerator must also be zero when $x = -2$, because the denominator is zero there.

$$(-2)^2 + 6(-2) + k = 0$$
$$-8 + k = 0$$
$$k = 8$$

Substitute $k = 8$ and factorise the numerator.

$$\lim_{x \to -2} \frac{x^2 + 6x + 8}{x + 2} = \lim_{x \to -2} \frac{(x + 2)(x + 4)}{x + 2} = \lim_{x \to -2} (x + 4)$$
$$= -2 + 4$$
$$= 2$$

Teacher answers & working

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S3

Find each tangent equation using a gradient from first principles.

Use first principles to find the equation of the tangent to $y = 2x^2 + 2$ at $x = -2$.

Answer:

$y = -8x - 6$

Working:

First find the gradient from first principles.

$$f'(-2) = \lim_{h \to 0} \frac{f(-2+h) - f(-2)}{h}$$

$$f(-2+h) = 2(-2+h)^2 + 2$$

$$f'(-2) = \lim_{h \to 0} \frac{(10 - 8h + 2h^2) - 10}{h}$$

$$= \lim_{h \to 0} (-8 + 2h)$$

$$= -8$$

Find the point on the curve: $f(-2) = 10$, so the point is $(-2, 10)$.

Use point-gradient form.

$$y - 10 = -8(x - -2)$$

$$y = -8x - 6$$

S4

Find the parameter value that makes each function continuous.

Find k so that $f(x) = \{2x - 3, \quad x3$ is continuous at $x = 3$.

Answer:

$k = 3$

Working:

For continuity, the left-hand limit, right-hand limit, and function value must all be equal.

$$\lim_{x \to 3^-} f(x) = 2(3) - 3$$

$$= 3$$

$$\lim_{x \to 3^+} f(x) = 3(3) - 6$$

$$= 3$$

The two one-sided limits are equal, so $\lim_{x \to 3} f(x) = 3$.

Set the function value equal to the limit: $k = 3$.

Teacher answers & working

Limits and First Principles · Parameters and Tangents

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EA1

Find the parameter so the limit exists, then evaluate the limit.

Find k so that $\lim_{x\to 2} \dfrac{x^2+2x+k}{x-2}$ exists, then evaluate the limit.

Answer:

$k=-8,\ \lim = 6$

Working:

For a finite limit, the numerator must also be zero when $x=2$, because the denominator is zero there.

$$\begin{aligned} 2^2+2(2)+k &= 0\\ 8+k &= 0\\ k &= -8 \end{aligned}$$

Substitute $k=-8$ and factorise the numerator.

$$\begin{aligned} \lim_{x\to 2}\frac{x^2+2x-8}{x-2} &= \lim_{x\to 2}\frac{(x-2)(x+4)}{(x-2)}\\ &= \lim_{x\to 2}(x+4)\\ &= 2+4\\ &= 6 \end{aligned}$$

EA2

Find each tangent equation using a gradient from first principles.

Use first principles to find the equation of the tangent to $y=x^2+4$ at $x=4$.

Answer:

$y=8x-12$

Working:

First find the gradient from first principles.

$$f'(4)=\lim_{h\to 0}\frac{f(4+h)-f(4)}{h}$$

$$f(4+h)=(4+h)^2+4$$

$$\begin{aligned} f'(4) &= \lim_{h\to 0}\frac{(20+8h+h^2)-20}{h}\\ &= \lim_{h\to 0}(8+h)\\ &= 8 \end{aligned}$$

Find the point on the curve: $f(4)=20$, so the point is $(4,20)$.

Use point-gradient form.

$$\begin{aligned} y-20 &= 8(x-4)\\ y &= 8x-12 \end{aligned}$$

Teacher answers & working

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EB1

Find the parameter value that makes each function continuous.

Find k so that $f(x) = \{2x + 1, \quad x0$ is continuous at $x = 0$.

Answer:

$k = 1$

Working:

For continuity, the left-hand limit, right-hand limit, and function value must all be equal.

$$\lim_{x \to 0^-} f(x) = 2(0) + 1$$
$$= 1$$
$$\lim_{x \to 0^+} f(x) = 2(0) + 1$$
$$= 1$$

The two one-sided limits are equal, so $\lim_{x \to 0} f(x) = 1$.

Set the function value equal to the limit: $k = 1$.